

Supporting Information

Hydrophobicity Does Not determine Water Slip Length: Insights from Slip Length Mapping

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I. Relation between Amplitude Decay and Damping Coefficient

The total damping coefficient, γ_{total} , acting on the probe can be expressed as the sum of the damping due to the probe sphere, γ_{tip} , and the background damping, γ_{bulk} :

$$\gamma_{\text{total}} = \gamma_{\text{tip}}(h, b_t, b_s) + \gamma_{\text{bulk}}. \quad (\text{S1})$$

where h is probe-substrate distance, and b_t and b_s are the slip lengths at the probe and substrate surfaces, respectively. The background damping γ_{bulk} originates from various sources, such as viscous drag on the cantilever beam and friction due to thermal fluctuations.¹ However, it remains nearly constant and is therefore treated as a constant here. Using the definition of the quality factor $Q(h) = k/(\omega\gamma_{\text{total}}(h))$, Eq. S1 can be rewritten as:

$$\frac{k}{\omega Q(h)} = \gamma_{\text{tip}}(h, b_t, b_s) + \gamma_{\text{bulk}}. \quad (\text{S2})$$

At sufficiently large $h \gg R$, the influence of the substrate becomes negligible and the hydrodynamic drag on the probe reduces to the Stokes resistance, $\gamma_{\text{tip}} = 6\pi\eta R$. Because this drag on a 300 nm spherical tip is orders of magnitude smaller than the viscous damping acting on the micrometer-scale cantilever, γ_{tip} becomes negligibly small compared to γ_{bulk} , leading to:

$$\gamma_{\text{total}} = \gamma_{\text{bulk}} = \frac{k}{\omega Q_{\text{bulk}}}. \quad (\text{S3})$$

Combining Eqs. S2 and S3 yields:

$$\frac{Q_{\text{bulk}}}{Q(h)} = \frac{\gamma_{\text{tip}}(h, b_t, b_s)}{\gamma_{\text{bulk}}} + 1. \quad (\text{S4})$$

In normal FM-AFM, the oscillation amplitude is stabilized by automatic gain control (AGC). In the present method, AGC was disabled and the cantilever was driven at a constant power. As a result, the amplitude decay directly reflects the damping, and the following relation

17 between the quality factor and the amplitude holds:

$$\frac{Q_{\text{bulk}}}{Q(h)} = \frac{A_{\text{bulk}}}{A(h)}. \quad (\text{S5})$$

18 Therefore, from Eqs. S4 and S5, the damping coefficient is related to the amplitude attenu-
19 ation ratio as follows:

$$\gamma_{\text{tip}}(h, b_t, b_s) = \gamma_{\text{bulk}} \left(\frac{A_{\text{bulk}}}{A(h)} - 1 \right). \quad (\text{S6})$$

20 **II. Comparison of Sensitivity between Conventional and Proposed** 21 **Methods**

22 We compared the sensitivity of slip length measurements between conventional contact-mode
23 AFM and the proposed FM-AFM method. In contact-mode AFM, the cantilever deflection
24 x is measured as a sensor voltage, and the relationship between the deflection sensor voltage
25 V_x and x is expressed using the sensitivity s as $x = sV_x$. Furthermore, to determine the
26 viscous drag in Eq. 3 from the restoring force of the cantilever, kx (k is the spring constant),
27 the following relation holds:

$$F = -ksV_x = -\frac{6\pi\eta R^2 \dot{h}}{h} f^*(h, b_t, b_s). \quad (\text{S7})$$

28 In contrast, in the FM-AFM method, the oscillation amplitude A is recorded as a sensor
29 voltage V_A , with the relationship $A = sV_A/\sqrt{2}$. From Eqs. 6 and 7 of the manuscript, the
30 following relation is obtained:

$$\frac{6\pi\eta R^2}{h} f^*(h, b_t, b_s) = \gamma_{\text{bulk}} \left(\frac{V_{A,\text{bulk}}}{V_A(h)} - 1 \right). \quad (\text{S8})$$

31 Here, the measurement sensitivity is defined as the change in signal voltage with respect to
32 the substrate slip length, $\partial V/\partial b_s$. Differentiating Eqs. S7 and S8 yields the sensitivities for

each method, S_{contact} and S_{FM} :

$$S_{\text{contact}} = \frac{\partial V_x}{\partial b_s} = -\frac{6\pi\eta R^2 \dot{h}}{ksh} \frac{\partial f^*}{\partial b_s}. \quad (\text{S9})$$

$$S_{\text{FM}} = \frac{\partial V_A}{\partial b_s} = -\frac{6\pi\eta R^2 V_{A,\text{bulk}}}{h\gamma_{\text{bulk}}} \left(1 + \frac{6\pi\eta R^2 f^*}{h\gamma_{\text{bulk}}}\right)^{-2} \frac{\partial f^*}{\partial b_s}. \quad (\text{S10})$$

From Eqs. S9 and S10, the ratio of sensitivities is given by:

$$\frac{S_{\text{FM}}}{S_{\text{contact}}} = \frac{V_{A,\text{bulk}} ks}{\gamma_{\text{bulk}} \dot{h}} \left(1 + \frac{6\pi\eta R^2 f^*}{h\gamma_{\text{bulk}}}\right)^{-2}. \quad (\text{S11})$$

The sensitivity ratio was calculated to be $S_{\text{FM}}/S_{\text{contact}} = 159$ using typical physical parameters in the measurements: $V_{A,\text{bulk}} = 0.2 \text{ V}$, $k = 20 \text{ N m}^{-1}$, $s = 10 \text{ nm V}^{-1}$, $\gamma_{\text{bulk}} = 2.3 \text{ } \mu\text{N s m}^{-1}$, $\dot{h} = 100 \text{ } \mu\text{m s}^{-1}$, $R = 300 \text{ nm}$, $\eta = 1 \text{ mPa s}$, $f^* = 0.63$, and $h = 10 \text{ nm}$. The f^* value corresponds to the slip length of $b_t = 0 \text{ nm}$ and $b_s = 10 \text{ nm}$, and the tip substrate distance $h = 10 \text{ nm}$ is chosen as a representative value among the range of distances used in the calculation. This indicates that the present method is 159 times more sensitive to slip length than the conventional approach. Consequently, slip length can be measured using much smaller colloidal probes than before.

III. Materials

A. Fabrication of Hydrophilic/Hydrophobic Composite Substrates

A thermally grown silica surface on a Si wafer was spin-coated with an electron-beam resist (ZEP520A, ZEON Corporation, Japan) at 5000 min^{-1} to form a 300 nm -thick film. Using an electron-beam lithography system (JSM-6360, JEOL Ltd., Japan / BEAM DRAW, Tokyo Technology Inc., Japan), square patterns of $1 \text{ } \mu\text{m} \times 1 \text{ } \mu\text{m}$ were patterned with a pitch of $2 \text{ } \mu\text{m}$, and the resist in the exposed areas was removed during development. Subsequently, a 5 nm -thick indium tin oxide (ITO) film was deposited by magnetron sputtering (MS-4B-3T, Cosmo System Co., Ltd., Japan). The electron-beam resist was then lifted off by rinsing

in dimethylacetamide for 5 min, leaving ITO patches on the silica substrate. The patterned ITO substrate was treated with O₂ plasma (PR500, Yamato Scientific Co., Japan) and then immersed in a 1 mmol L⁻¹ ethanol solution of 1H,1H,2H,2H-perfluorooctanephosphonic acid (FOPA) for 1 h to form a self-assembled monolayer (SAM). Since FOPA SAMs are easily hydrolyzed on silica but remain stable on ITO surfaces and maintain hydrophobicity in water for several hours,^{2,3} the fabricated composite substrate was immersed in water for 3 h to selectively remove FOPA from the silica surface. Contact angles of the hydrophilic and hydrophobic regions were evaluated immediately before the FM-AFM measurements using unpatterned reference substrates that underwent the corresponding O₂-plasma treatment, FOPA/ethanol-solution immersion, and subsequent water-immersion steps used to prepare the composite substrate.

B. Fabrication of Teflon Substrates

Silica substrates cut into 1 cm×1 cm were ultrasonically cleaned in acetone (US-1KS, SND Co., Ltd., Japan) and treated with O₂ plasma. After baking at 120 °C for 5 min to remove surface moisture, the substrates were immersed for 1 h in a 0.05 % (w/w) toluene solution of FDTS to form an FDTS monolayer. On top of the FDTS-treated substrate, a 0.6 % (w/w) solution of Teflon AF1600X (DuPont Inc., Wilmington, DE, USA) in FC-770 (3M, St. Paul, MN, USA) was dropped, spin-coated at 4000 min⁻¹ for 1 min, and baked at 175 °C for 10 min. The FDTS underlayer was used to improve the adhesion of the Teflon coating.⁴

Table S1: Literature values of contact angle and slip length for various solid-liquid interfaces

Author	Method	Substrate	Contact Angle θ (°)	Slip length (nm)
Bonaccorso et al. 2002 ⁵	AFM	Mica	0	8.5 ± 0.5
Vinogradova & Yakubov 2003 ⁶	AFM	Silica	0	0 ± 1
		Polystyrene	86–92	4 ± 1
Cho et al. 2004 ⁷	AFM	OTS	97.5	30
Cottin-Bizonne et al. 2005 ⁸	AFM	OTS-Pyrex	105	19 ± 2
Maali et al. 2008 ⁹	AFM	HOPG	74	8 ± 2
		Mica	0	2
Bouzigues et al. 2008 ¹⁰	TIRF-PIV	Silica	20	0 ± 10
		OTS	95	38 ± 6
		Silica	20	3 ± 7
		OTS	95	29 ± 11
Cottin-Bizonne et al. 2008 ¹¹	SFA	OTS-Pyrex	105	17 ± 2
		Pyrex	3	0
		DPPC monolayer	95	10 ± 2
Lasne et al. 2008 ¹²	TIRF-PIV	OTS	90	45 ± 15
		Silica	0	0
Bhushan et al. 2009 ¹³	AFM	Mica	0	0
		n-hexatriacontane	91 ± 2	43 ± 10
		Lotus wax	167 ± 0.7	236 ± 18
Zhu et al. 2011 ¹⁴	AFM	OTS	112	26 ± 3
Jing & Bhushan 2013 ¹⁵	AFM	OTS	106 ± 2	28 ± 5
Ortiz-Young et al. 2013 ¹⁶	AFM	Mica	4 ± 3	0
		GO	48 ± 3	0.24 ± 0.38
		DLC	66 ± 6	0.55 ± 1.37
		Si	76 ± 4	1 ± 1.7
		HOPG	85 ± 4	12 ± 3.3
		DLC	75–100	2.4
Xue et al. 2015 ¹⁷	QCM	MUOH	40–80	1.9
		MUOH	0–60	1.2
		MUOH	0–50	0.7
		MUOH	0–50	0.7
Ahmad et al. 2015 ¹⁸	AFM	Si	0	0
		OTS	108 ± 3	0
Radha et al. 2016 ¹⁹	Channel	Graphene	55–85	60
Zhang et al. 2021 ²⁰	AFM	Mica	10	0
		Talc	75	95
Keerthi et al. 2021 ²¹	Channel	Graphene	62	60
		hBN	62	1
Li et al. 2022 ²²	AFM	OTS	100	13.4 ± 3
		SiO ₂	24.9	0.9 ± 1.2
		HOPG	75 ± 5	4.3 ± 3.5
Chen et al. 2022 ²³	Channel	Graphene	83.7	40 ± 4
		N/A	83.7	33 ± 3
Han et al. 2025 ²⁴	AFM	SiO ₂	41.2 ± 0.7	4.3
		APTES	68.2 ± 1.9	9.4
		OTS	108.5 ± 0.1	29.1
		FDTES	125.7 ± 1.8	65.3

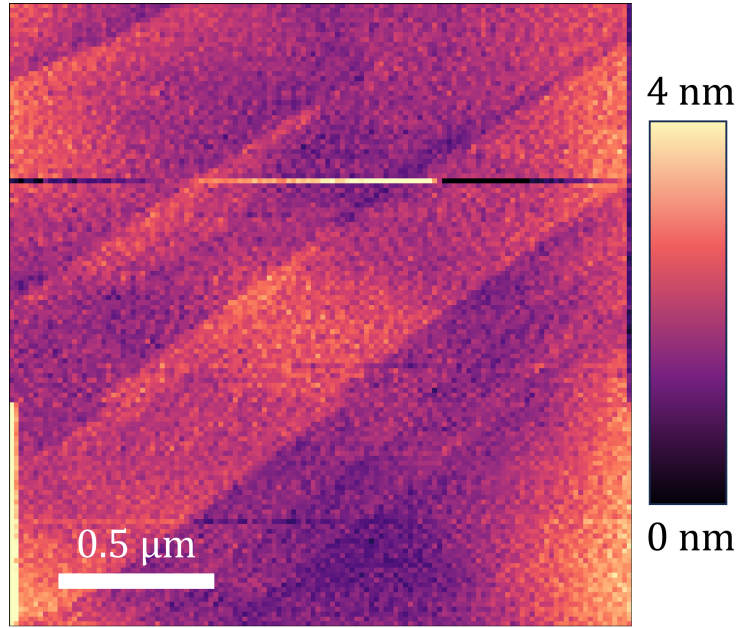


Figure S1: Height image acquired while the tip was strongly pressed against the surface, showing disappearance of the bubble-like feature and exposure of the underlying HOPG substrate.

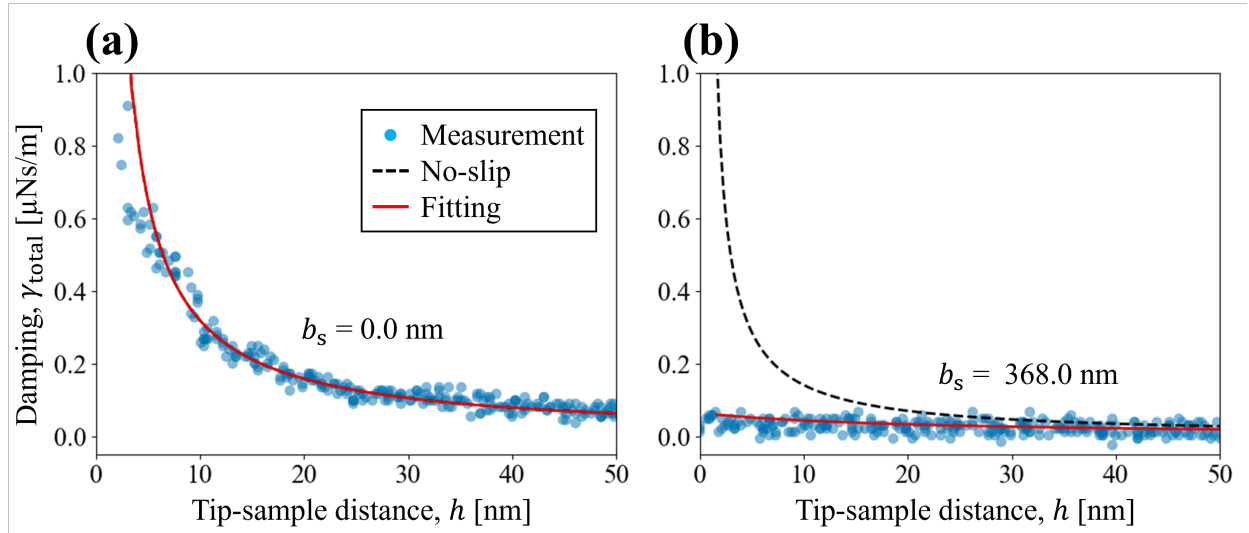


Figure S2: Representative damping-coefficient fitting curves for (a) an atmospheric-plasma-treated DLC substrate and (b) a location above a nanobubble. The slip lengths shown in the panels were obtained from these representative single-point fits and therefore do not necessarily coincide with the mean values reported in the main text: 0.0 ± 1.2 nm for DLC and 345.3 ± 23.7 nm for nanobubbles. The mean absolute errors were $0.016 \mu\text{Ns m}^{-1}$ and $0.014 \mu\text{Ns m}^{-1}$ for (a) and (b), respectively.

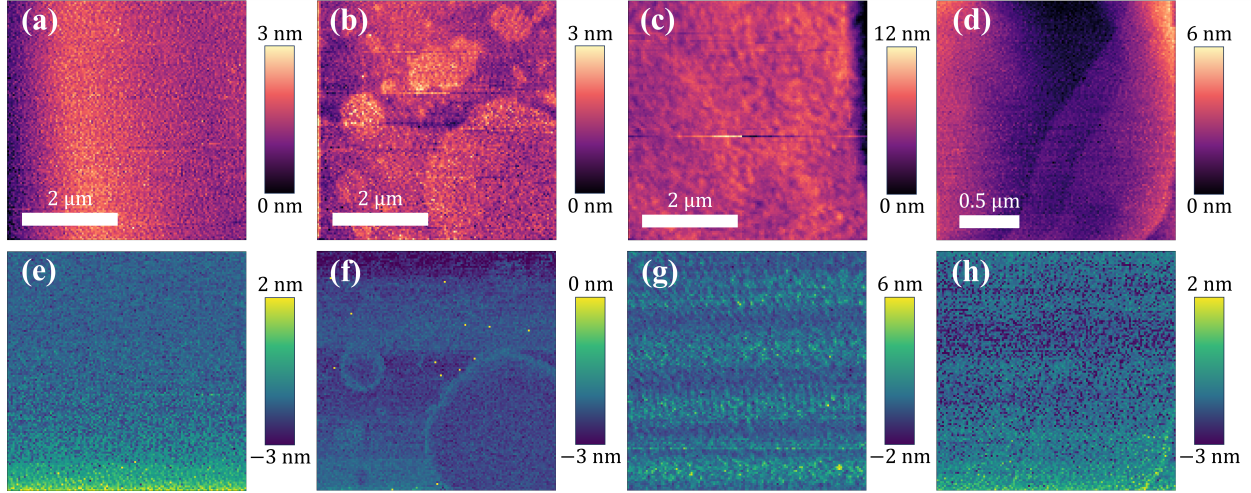


Figure S3: (a–d) Topography and (e–h) corresponding slip-length maps of (a, e) mica, (b, f) FDTS, (c, g) Teflon, and (d, h) HOPG. The R_a values were 0.27 nm, 0.29 nm, 0.64 nm, and 0.35 nm for each surface, respectively. Only the result for HOPG (d, h) was obtained in a 10 mmol L⁻¹ KCl solution and the others were in DI water.

Table S2: Measured slip lengths and values predicted by Eq. 2 for the substrates investigated in this study.

Substrate	Contact Angle (°)	b predicted by $C = 0.41$ nm (nm)	b predicted by $C = 6.0$ nm (nm)	Measured b (nm)
Mica	< 3	~ 0.10	~ 1.50	-1.7 ± 0.3
Oxygen plasma-treated silica	28.1	0.12	1.69	-0.7 ± 0.6
FDTS	97.9	0.55	8.06	-1.7 ± 0.2
FOPA	105.0	0.75	10.92	-0.9 ± 0.6
Teflon	111.1	1.00	14.65	0.8 ± 0.8
HOPG (DI water)	59.0	0.18	2.61	43.2 ± 5.8
HOPG (10 mM KCl aq.)	62.8	0.19	2.83	-0.1 ± 0.7
HOPG (10 mM NaCl aq.)	57.0	0.17	2.51	-2.3 ± 0.6

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